

Problem Set 70

Practicing Extrema

April 30, 2026

Problem 2

The graph of $f'(x)$, shown below on the interval $[0, 10]$, consists of a semicircle of radius 2 and a line segment meeting at the point $(4, 0)$. The line segment has right endpoint $(10, 2)$.

(a) For what value of x , $0 < x < 10$, does f have a local minimum? Justify your answer.

$f(x)$ has a local minimum at $x = 4$ as $f'(x)$ changes from negative to positive at $x = 4$.

(b) For what value of x , $0 \leq x \leq 10$, does f have an absolute maximum? Justify your answer.

As no other critical points are present, one attains the absolute maximum of the function by analyzing the end points.

$$\begin{aligned} f(10) &= f(0) + \int_0^{10} f'(x) dx \\ &= f(0) - \pi \cdot 2^2 \cdot \frac{1}{2} + \frac{1}{2} \cdot 2 \cdot 6 \\ &= f(0) + 6 - 2\pi \\ f(10) &< f(0) \implies \boxed{x = 0} \end{aligned}$$

Hence, $f(x)$ has an absolute maximum at $x = 0$.

Problem 4

The velocity of a particle moving on the x -axis is given by $v(t) = t^3 - 6t^2$ for the time interval $0 \leq t \leq 10$.

(a) When is the particle farthest to the left?

$$\begin{aligned} v(t) &= x'(t) \\ v(t) = 0 &\implies t^2(t - 6) = 0 \\ t &= 6 \end{aligned}$$

As $v(t)$ changes from negative to positive at $t = 6$, the particle is farthest to the left at $t = 6$.

(b) When is the velocity of the particle increasing the fastest?

$$\begin{aligned} v'(t) &= 3t^2 - 12t \\ v''(t) &= 6t - 12 \\ v''(t) = 0 &\implies t = 2 \end{aligned}$$

$$\begin{aligned}v'(1) &= -9 \\v'(0) &= 0 \\v'(10) &= 280\end{aligned}$$

The velocity increases the fastest at $t = 10$.

Problem 5

Let $\frac{dy}{dx} = 2x + y$.

(a) How do we know that $(1, -2)$ is a critical point for the function $y = f(x)$?

$$\frac{dy}{dx} = 0 \implies 2x + y = 0$$

$2 \cdot 1 + (-2) = 0$

(b) Is the point $(1, -2)$ a relative maximum point, relative minimum point, or neither? Justify your answer.

$$\frac{d^2y}{dx^2} = 2 + \frac{dy}{dx}$$

$$x = 1 \implies \frac{d^2y}{dx^2} = 2 > 0$$

As $f'(x) = 0$ and $f''(x) > 0$, $f(x)$ has a relative minimum point at $x = 1$.

Problem 6

Let f be a differentiable function with $f(4) = 3$. On the interval $0 \leq x \leq 7$, the graph of f' , the derivative of f , consists of a semicircle and two line segments, as shown in the figure below.

(a) Find the x -coordinates of all points of inflection on the graph of f for $0 < x < 7$. Justify your answer.

$$f''(x) = 0 \implies \boxed{x = 2, 6}$$

$f''(x)$ changes sign at $x = 2, 6$.

(b) Let g be the function defined by $g(x) = f(x) - x$. On what intervals, if any, is g decreasing for $0 \leq x \leq 7$? Show the analysis that leads to your answer.

$$\begin{aligned}g'(x) &= f'(x) - 1 \\g'(x) &< 0, \quad x \in [0, 5)\end{aligned}$$

$g(x)$ decreases on $[0, 5)$ as $g'(x) < 0$ on this interval.

(c) For the function g defined in part (b), find the absolute minimum value on the interval $0 \leq x \leq 7$. Justify your answer.

$$g'(x) = 0 \implies x = 5 \text{ or } 7$$

As $g'(x)$ changes from negative to positive at $x = 5$, $g(x)$ has an absolute minimum at $x = 5$.

Problem 7

The height of a tree at time t is given by a twice-differentiable function H , where $H(t)$ is measured in meters and t is measured in years. Selected values of $H(t)$ are given in the table below.

(a) Use the data in the table to estimate $H'(6)$. Using correct units, interpret the meaning of $H'(6)$ in the context of the problem.

$$H'(6) \approx \frac{H(7) - H(5)}{7 - 5} \approx \frac{11 - 6}{7 - 5} \approx \boxed{\frac{5}{2} \text{ meters/years}}$$

$H'(6)$ represents the rate of growth of the tree at year 6.

(b) Explain why there must be at least one time t , for $2 < t < 10$, such that $H'(t) = 2$.

As $H(t)$ is twice-differentiable, $H(t)$ is continuous on $[3, 5]$ and differentiable on $(3, 5)$.

By the mean value theorem,

$$\boxed{\exists c \in (3, 5) \text{ s.t. } f'(c) = \frac{f(5) - f(3)}{5 - 3} = 2}$$

Problem 9

The function f is differentiable on the closed interval $[-6, 5]$ and satisfies $f(-2) = 7$. The graph of f' , the derivative of f , consists of a semicircle and the line segments, as shown in the figure below.

(a) On what intervals is f increasing? Justify your answer.

$$f'(x) > 0, \quad x \in [-6, -2) \cup (2, 5)$$

$f(x)$ is increasing on $[-6, -2)$ and $(2, 5)$ as $f'(x) > 0$ on these intervals.

(b) Find the absolute minimum value of f on the closed interval $[-6, 5]$. Justify your answer.

$$f'(x) = 0 \implies x = -2, 2$$

As $f'(x)$ changes from negative to positive at $x = 2$, $f(x)$ has an absolute minimum at $x = 2$.

(c) For each of $f''(5)$ and $f''(3)$, find the value or explain why it doesn't exist.

$$\begin{aligned} f''(5^+) \text{ DNE} &\implies \boxed{f'(5) \text{ DNE}} \\ f''(3^-) = 2, f''(3^+) = -1 \\ f''(3^-) \neq f''(3^+) &\implies \boxed{f'(3) \text{ DNE}} \\ f'(5) &: \text{Endpoint Discontinuity} \\ f'(3) &: \text{Corner Discontinuity} \end{aligned}$$

Problem 10

Let f be the function given by $f(x) = \frac{\ln x}{x}$ for all $x > 0$. The derivative of f is given by

$$f'(x) = \frac{1 - \ln x}{x^2}.$$

(a) Write an equation for the line tangent to the graph of f at $x = e^2$.

$$\begin{aligned} f'(e^2) &= \frac{1 - \ln e^2}{e^4} = -\frac{1}{e^4} \\ f(e^2) &= \frac{\ln e^2}{e^2} = \frac{2}{e^2} \\ \boxed{y - \frac{2}{e^2} = -\frac{1}{e^4}(x - e^2)} \end{aligned}$$

(b) Find the x -coordinate of the critical point of f . Determine whether this point is a relative minimum, a relative maximum, or neither for the function f . Justify your answer.

$$\begin{aligned} f'(x) &= \frac{1 - \ln x}{x^2} = 0 \\ 1 - \ln x &= 0 \implies \boxed{x = e} \end{aligned}$$

As $f'(x)$ changes from positive to negative at $x = e$, $f(x)$ has a relative maximum at $x = e$.

(c) The graph of the function f has exactly one point of inflection. Find the x -coordinate of this point.

$$\begin{aligned} f''(x) &= \frac{2 \ln x - 3}{x^3} \\ f''(x) &= 0 \implies 2 \ln x = 3 \\ \ln x &= \frac{3}{2} \implies \boxed{x = e^{\frac{3}{2}}} \end{aligned}$$

(d) Find $\lim_{x \rightarrow 0^+} f(x)$.

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\ln x}{x} \rightarrow \boxed{-\infty}$$